IF-ELSE STATEMENT

CONDITIONAL STATEMENTS

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Flowchart

if Statement

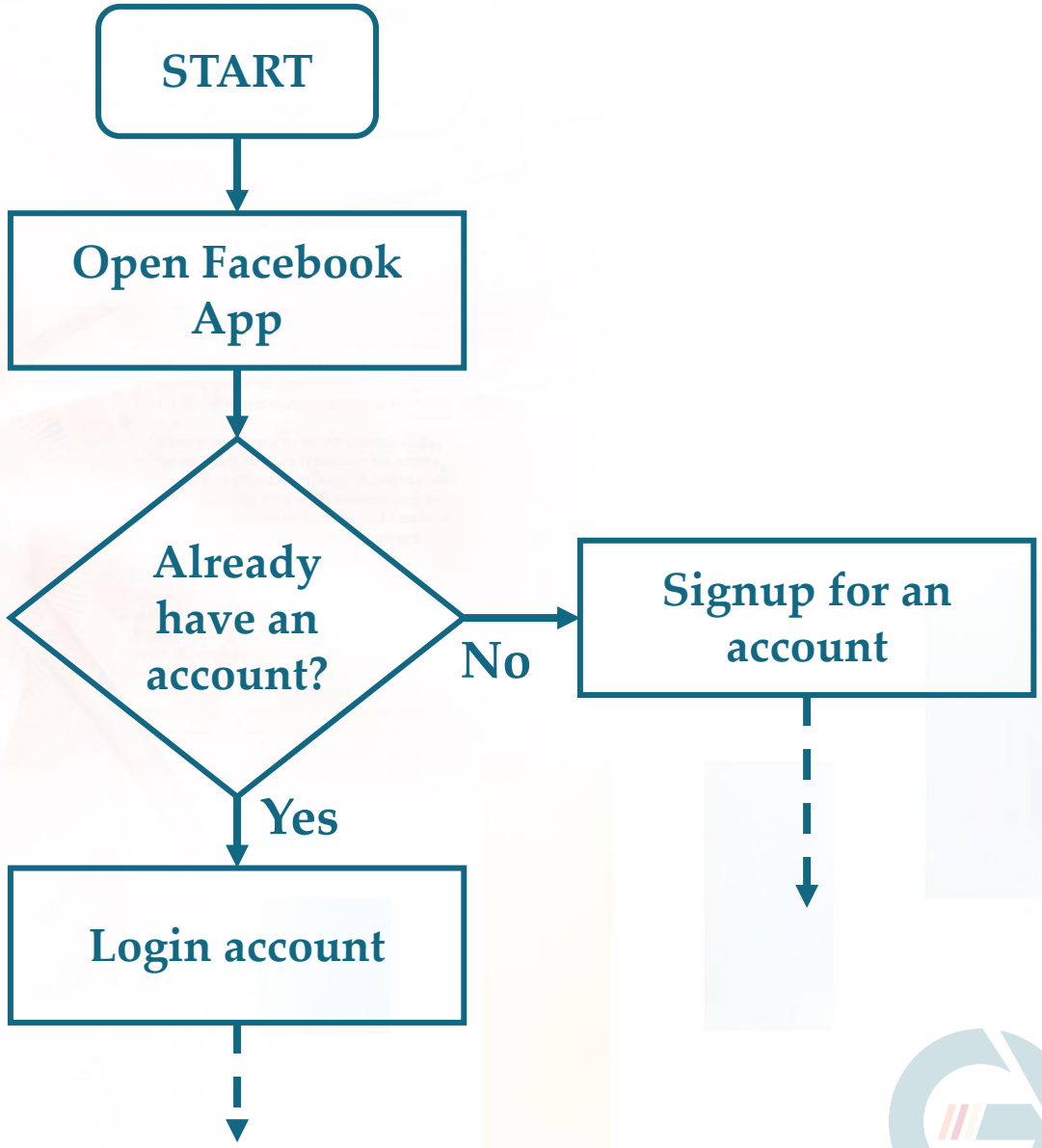
if-else Statement

if-else if-else Statement

FLOWCHART

FLOWCHART

A diagram that illustrates the **steps**, **sequences**, and decisions of a process or workflow



FLOWCHART SYMBOLS

Start/End

Process

Input/Output

Decision

IF STATEMENT

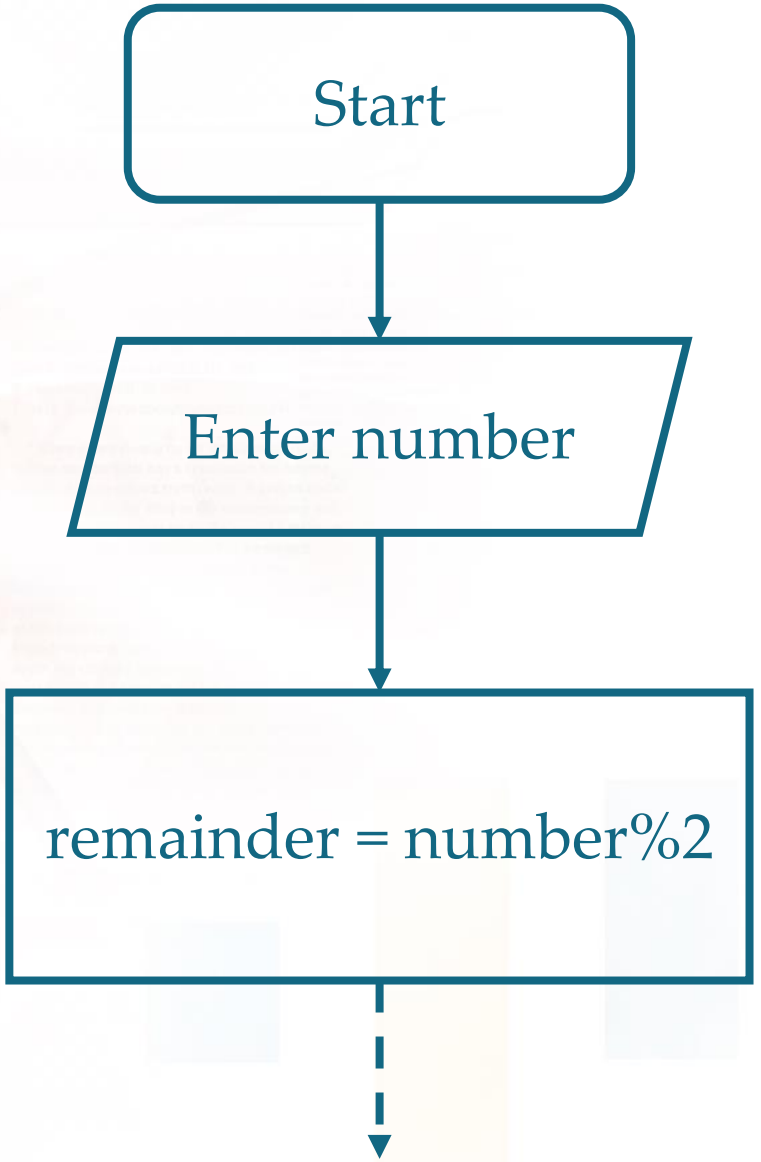
IF STATEMENT

```
if (condition) {
    statement;
}
```

The if conditional statement is used to execute a statement or block, if, and only if, the **condition** is **true**

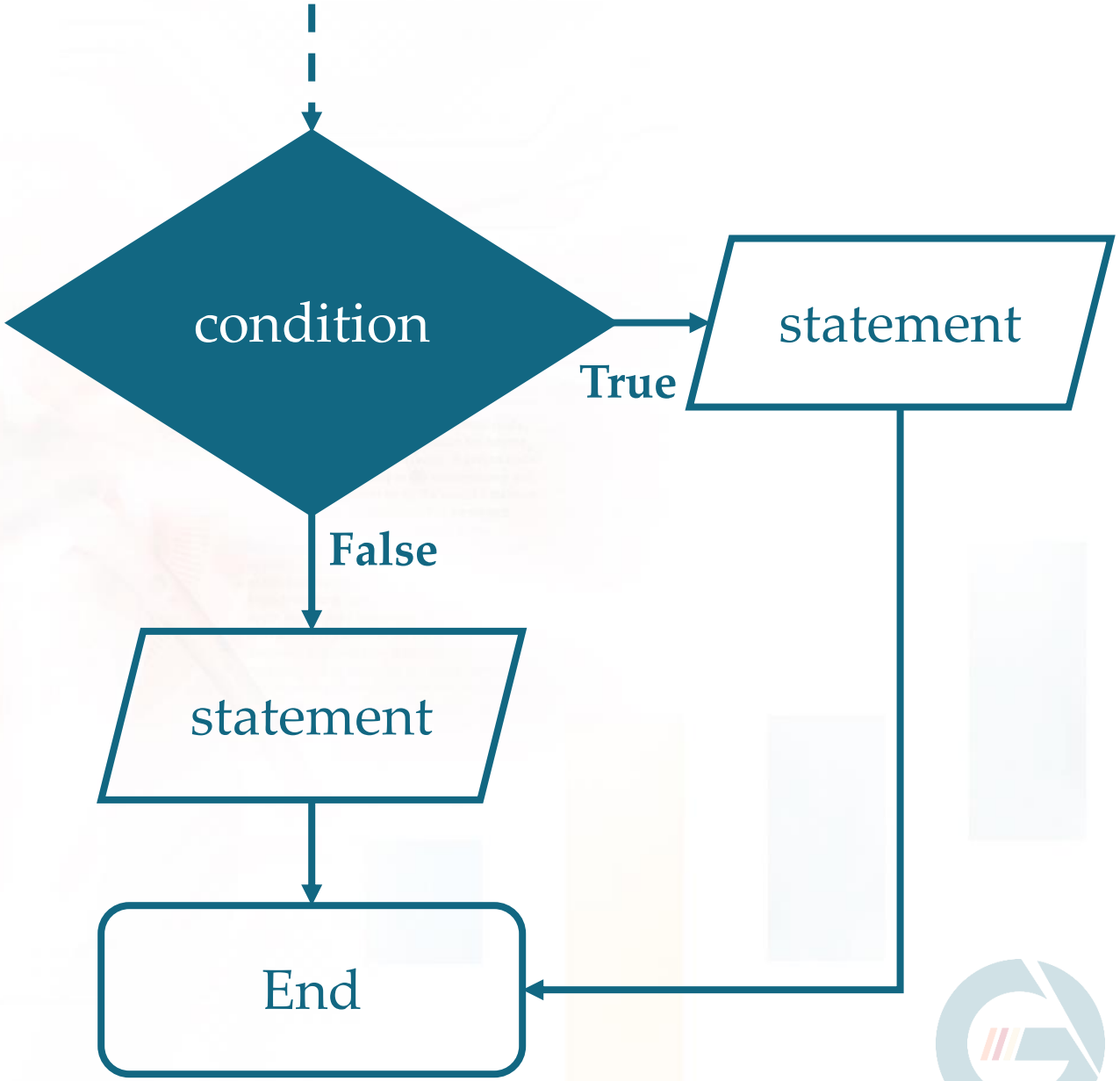
IF STATEMENT

```
if (condition) {
    statement;
}
```

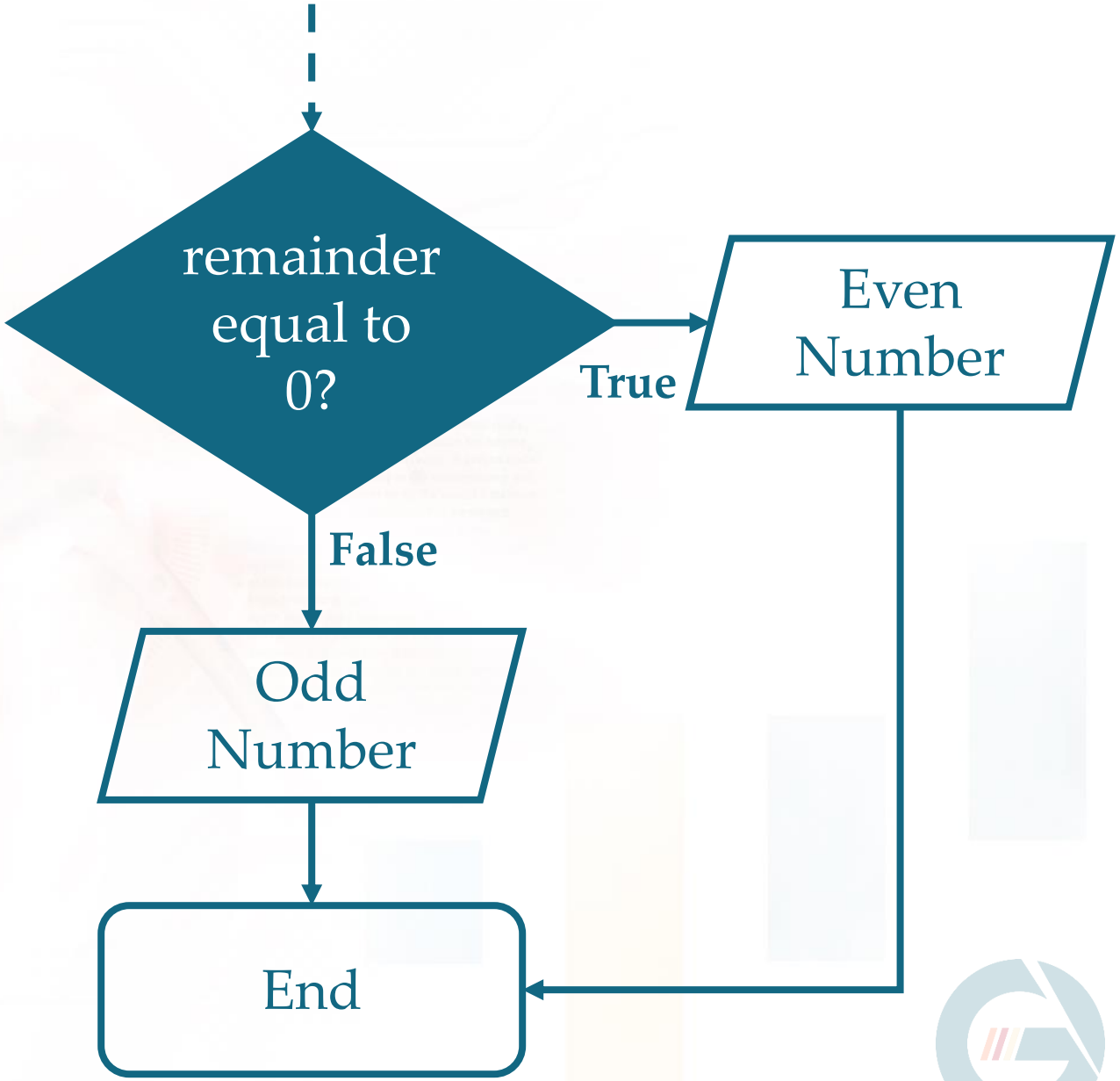


IF STATEMENT

```
if (condition) {
    statement;
}
```



```
if (remainder == 0) {
    cout<< "Even Number";
}
```



IF-ELSE STATEMENT

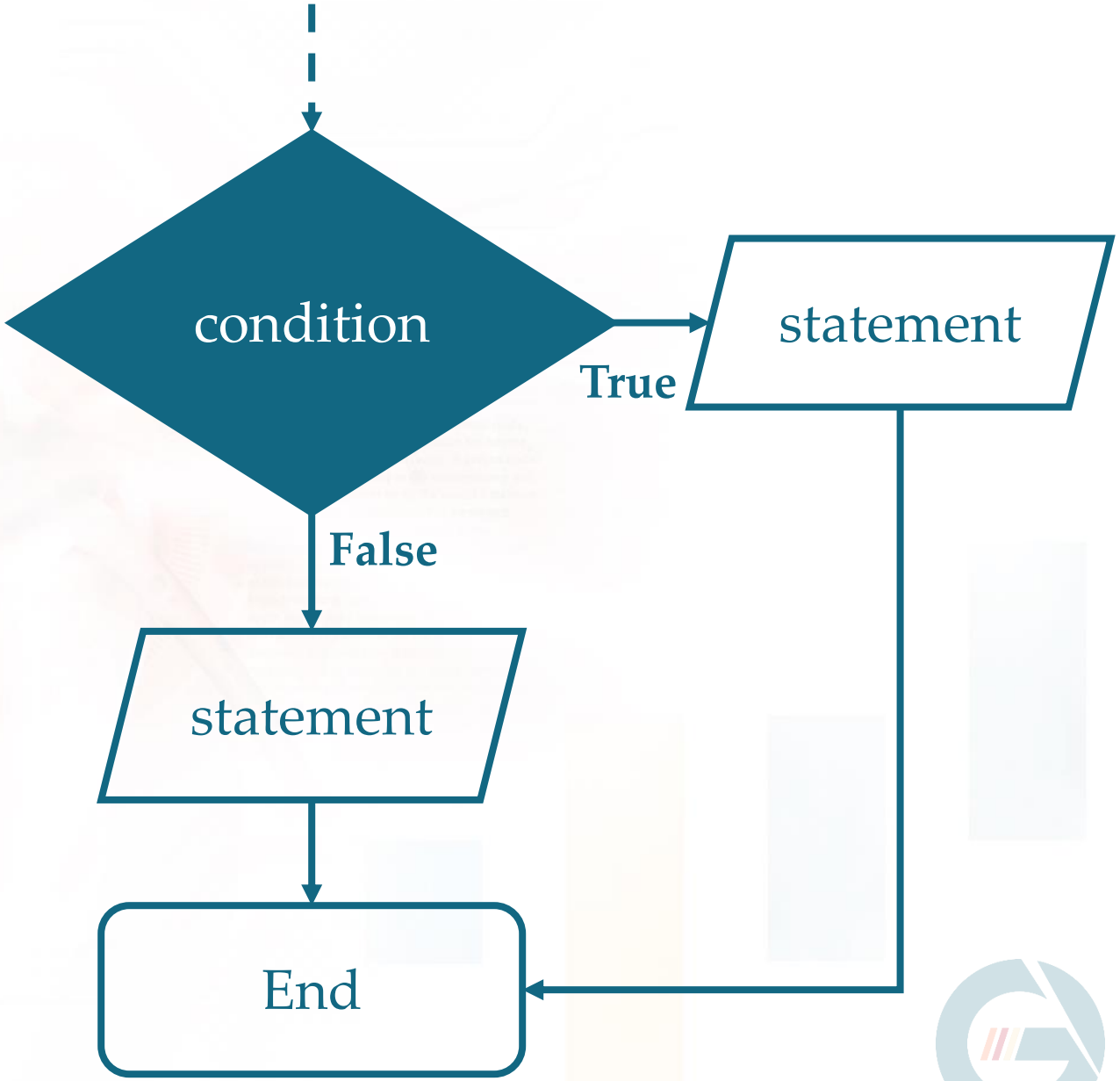
IF-ELSE STATEMENT

```
if (condition) {
    statement;
}
else{
    statement;
}
```

The if-else statement executes one block of code if the **condition** is **true** and another block if it is **false**

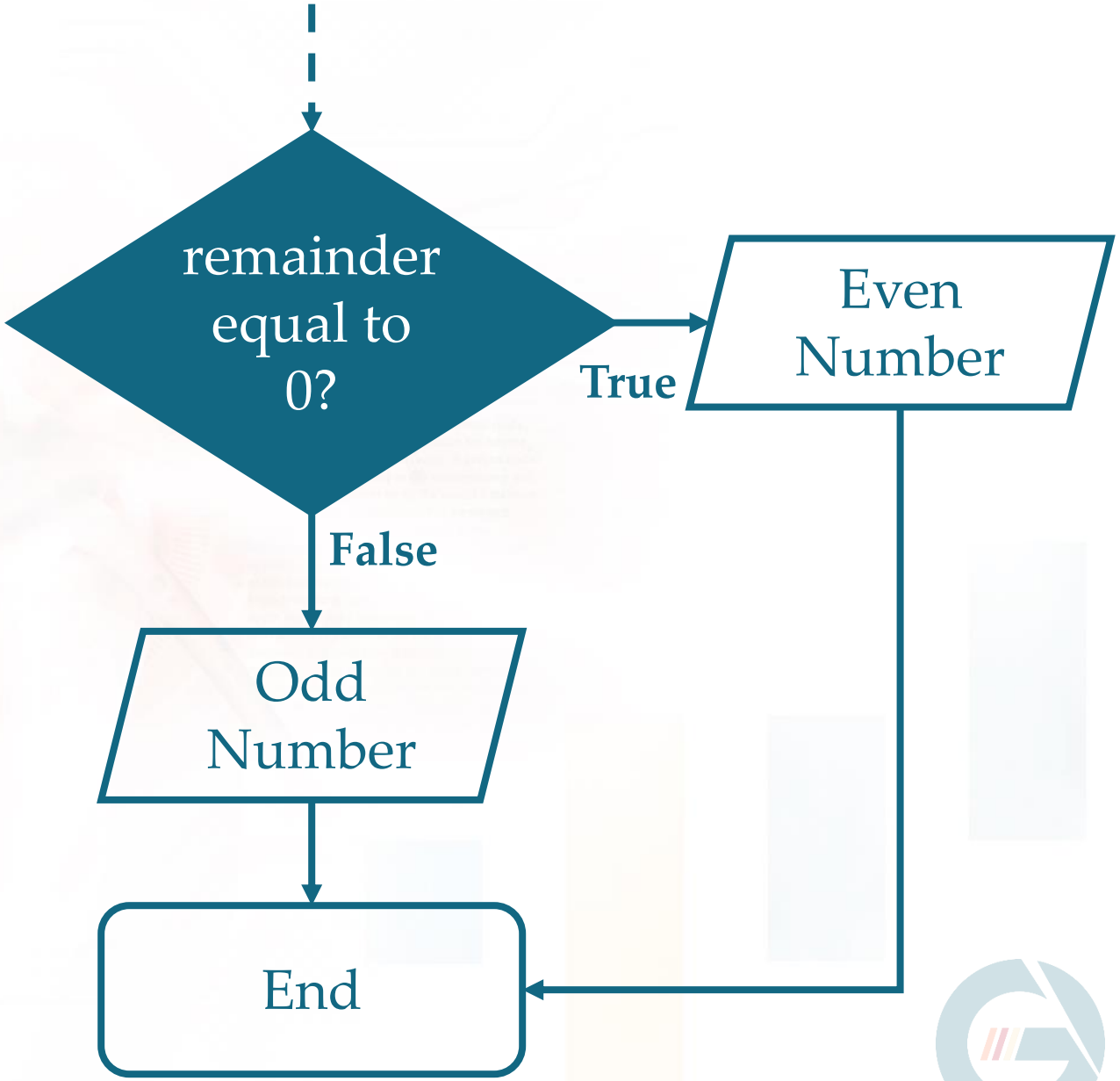
IF STATEMENT

```
if (condition) {
    statement;
}
else{
    statement;
}
```



IF-ELSE STATEMENT

```
if (remainder == 0) {
    cout<< "Even Number";
}
else{
    cout<<"Odd Number";
}
```



IF vs IF-ELSE STATEMENT

```
remainder = 1;
if (remainder == 0) {
    cout<< "Even Number";
}
if (remainder != 0){
    cout<<"Odd Number";
}
```

✗ $1ms$

✓ $1ms$

Runtime = 2ms

SAME

```
remainder = 1;
if (remainder == 0) {
    cout<< "Even Number";
}
else{
    cout<<"Odd Number";
}
```

✗ $1ms$

✓ $1ms$

Runtime = 2ms

SAME

IF vs IF-ELSE STATEMENT

```
remainder = 0;
if (remainder == 0) {
    cout<< "Even Number";
}
if (remainder != 0){
    cout<<"Odd Number";
}
```

$1ms$

$1ms$

Runtime = 2ms

SLOWER

```
remainder = 0;
if (remainder == 0) {
    cout<< "Even Number";
}
else{
    cout<<"Odd Number";
}
```

$1ms$

skip

Runtime = 1ms

FASTER

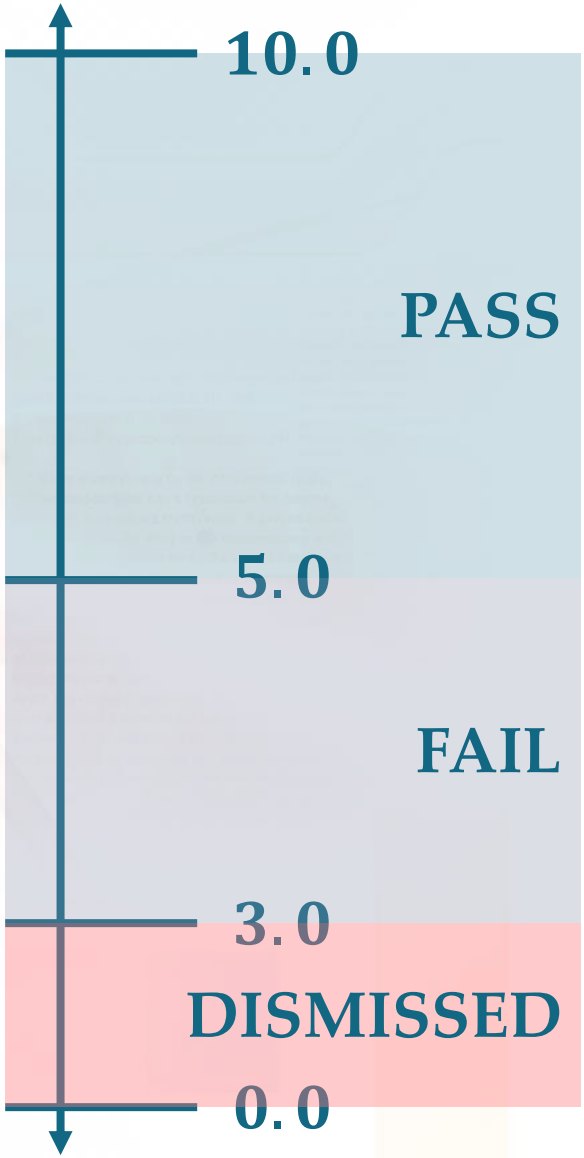
IF-ELSE IF-ELSE STATEMENT

IF-ELSE IF-ELSE

```
if (condition) {
    statement;
}
else if (condition) {
    statement;
}
else {
    statement;
}
```

The if-else if-else statement check **multiple conditions** and execute different blocks of code based on the evaluation of those conditions.

```
if (condition) {
    statement;
}
else if (condition) {
    statement;
}
else {
    statement;
}
```

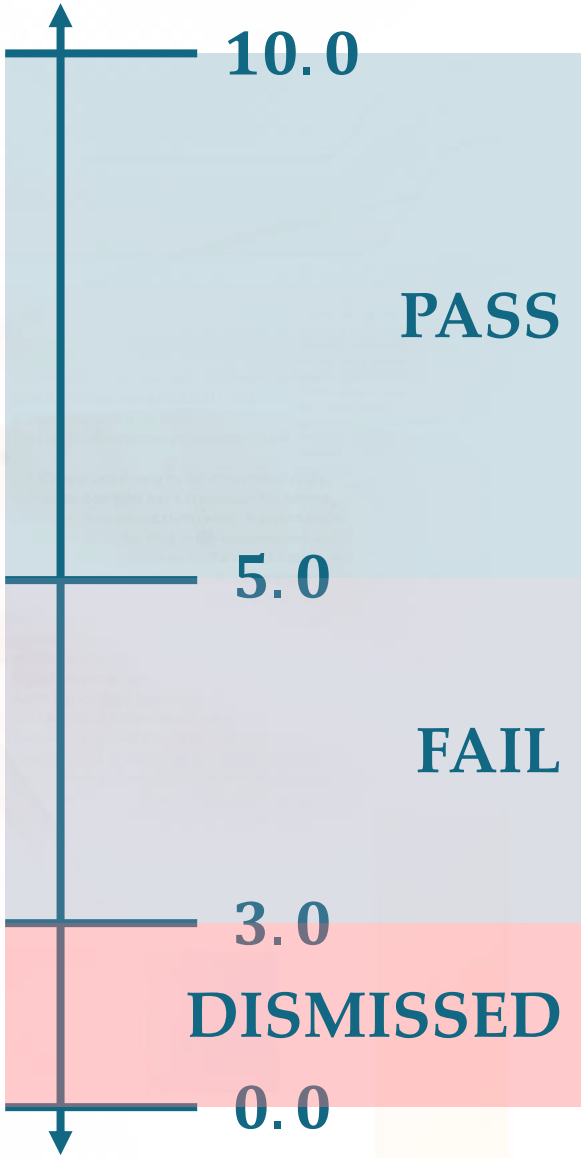


```
grade =  6.3;
if (grade >= 5.0) {
    cout << "PASS";
}
else if (grade <5.0 && grade >= 3.0) {
    cout << "FAIL";
}
else {
    cout << "DISMISSED";
}
```

1ms

skip

skip

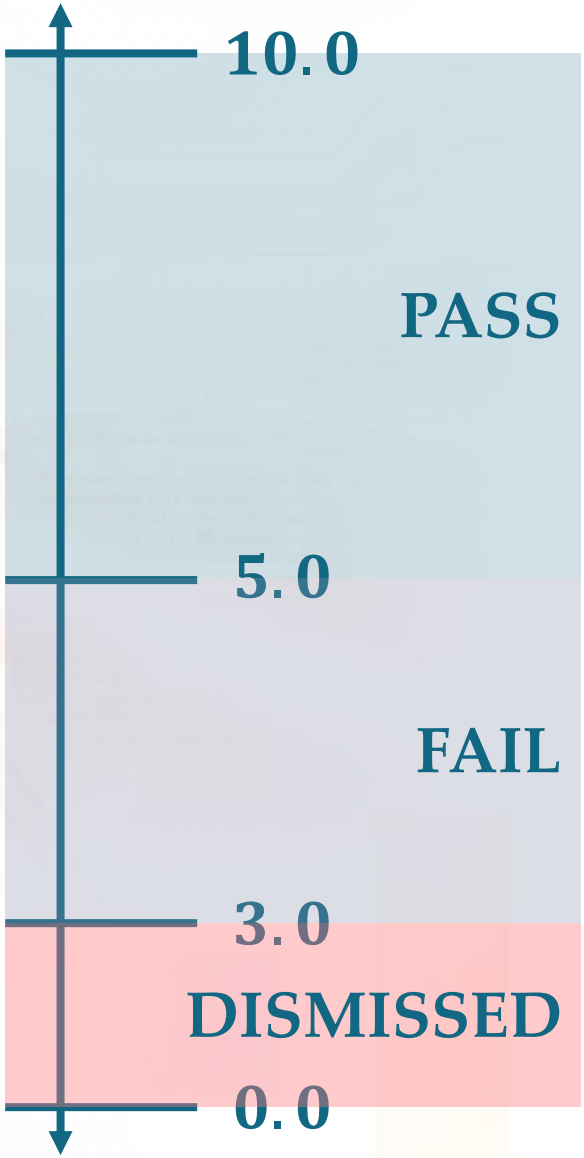


```
grade =  4.5;
if (grade >= 5.0) {
    cout << "PASS";
}
else if (grade <5.0 && grade >= 3.0) {
    cout << "FAIL";
}
else {
    cout << "DISMISSED";
}
```

1ms

1ms

skip

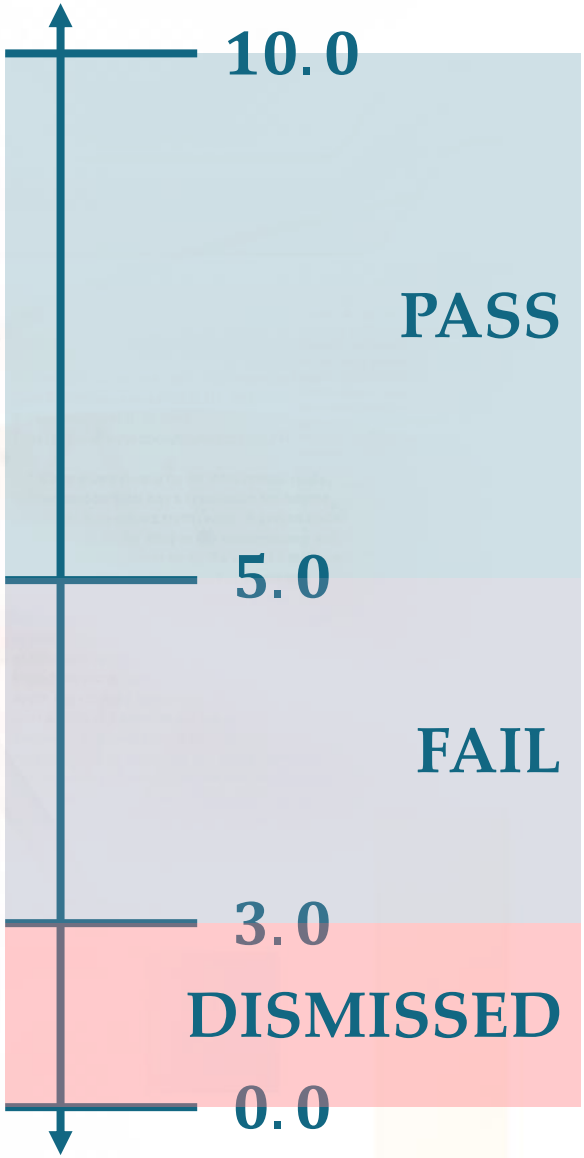


```
grade =  2.9;
if (grade >= 5.0) {
    cout << "PASS";
}
else if (grade <5.0 && grade >= 3.0) {
    cout << "FAIL";
}
else {
    cout << "DISMISSED";
}
```

$1ms$

$1ms$

$1ms$



```
#include <iostream>
#include <string>
using namespace std;
int main( ){
    // database
    string user_1 = “Jason”;
    string pass_1 = “Red”;
    string user_2 = “Zack”;
    string pass_2 = “Black”;
    string user_3 = “Kimberly”;
    string pass_3 = “Pink”;
```

```
// variables
string user_name = “name”;
string password = “pass”;

// login account
cout<<“Enter user name: “;
getline(cin, user_name);
cout<<“Enter password: “;
getline(cin, password);
```

```
// verify account
if (username == user_1 && password == pass_1){
        cout<<"Welcome Back "<<username"<<"!";
}
else if (username == user_2 && password == pass_2){
        cout<<"Welcome Back "<<username"<<"!";
}
else if (username == user_3 && password == pass_3){
        cout<<"Welcome Back "<<username"<<"!";
}
```

```
    else {
        cout<<"Invalid username or password!";
    }

    return 0;
}
```

LABORATORY